SUPPORTING ADULT GUIDANCE

Percentage change

Percentages, Ratio, Growth and Bounds · Lesson 4. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

In the main activity, the Before-and-After Board, learners set before and after values for a price, a village population or a journey time. The before bar is fixed at 100%; the change shows as a green piece (rise) or a dashed red gap (fall). The readout gives the change, then the change over the original as a stacked fraction × 100, and the multiplier (new over original). Every percentage is exact: when it is not a terminating decimal it is shown as an exact mixed number (for example 33 1/3%). Swap before and after shows +25% becoming –20%; Compare two changes puts two boards side by side. Seven questions and a stretch follow, then quick changes, true-or-not statements and briefs to meet.

Driving Question: *Change compared with what?*

The mathematics, in plain terms

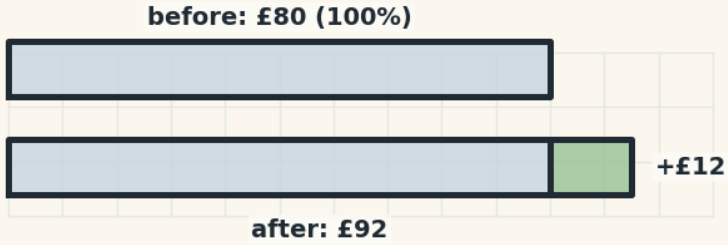
Percentage change = 100(new – original)/original. It is asymmetric: a change from a to b and back has percentages 100(b – a)/a and 100(a – b)/b, which differ unless a = b. Equivalently, the multiplier b/a has reciprocal a/b, and (1 + p)(1 – q) = 1 gives q = p/(1 + p): +25% is undone by –20%.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — A percentage increase

A jacket cost £80 last year and costs £92 now. What is the percentage increase?

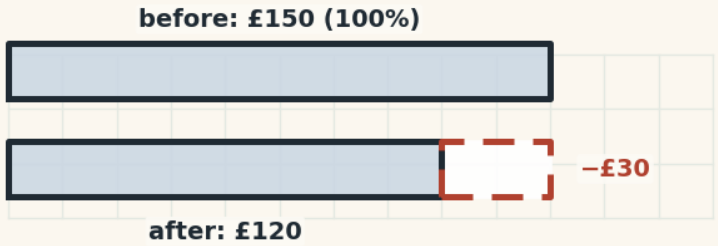


1. The change	£92 – £80 = £12
2. Compare with the original	$\frac{12}{80} = 0.15$
3. As a percentage	0.15 × 100 = 15%
4. Check with the multiplier	$\frac{92}{80} = 1.15$

A 15% increase.

Example 2 — A percentage decrease

A phone falls in price from £150 to £120. What is the percentage decrease?



1. The change	£150 – £120 = £30
2. Compare with the original, £150	$\frac{30}{150} = 0.2$
3. As a percentage	20%

A 20% decrease (not 25%, which would compare with the new price, £120).

See the original, then calculate

The before bar stays fixed at 100%, so learners see the change measured against it before they divide. Swapping and comparing make the role of the original visible, which is the source of the most common error (dividing by the new value).

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	The Before-and-After Board: one change, swapping, and comparing two. Seven questions. Stretch: +25% versus –20%.	~16 min
Movement	Quick changes.	~6 min
Reflection	True or not? Original, new and comparisons.	~8 min
Creativity	Make the change: six briefs.	~10 min
Rest	Where you start shapes how far you have come.	~2 min
Nutrition	Practical: judging claims in the news and adverts.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Dividing by the new value instead of the original.
Ask: “Which value did it start from?”

Giving the change in pounds as the answer.
Ask: “What is that change as a percentage of where it started?”

Assuming equal pound changes are equal percentage changes.
Ask: “Compared with what, in each case?”

Expecting a rise and an equal fall to be the same percentage.
Ask: “What is the original going up? And coming back down?”

Forgetting to say whether it is an increase or a decrease.
Ask: “Did it go up or down?”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: percentages as multipliers (Lesson 3) and one number as a percentage of another. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner says a fall from £250 to £200 is a 25% decrease. The aim is a question that brings back the original.

- “What was the price before the change?”

- “So the £50 is part of which amount?”
- “What is 50 out of 250?”
- “Where did the 25% come from?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Why are some percentages shown as fractions, like 33 1/3%?
So that every value on screen is exact. Learners can round sensibly in their own answers; the lesson accepts equivalent forms.

Should a decrease be written as a negative?
Either is accepted: “a 20% decrease” or “–20%”.

Is the stretch required?
No. The secure core is dividing the change by the right original and multiplying by 100.

What comes next?
Lesson 5: reverse percentages, working back to the original from the new value.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.

Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.